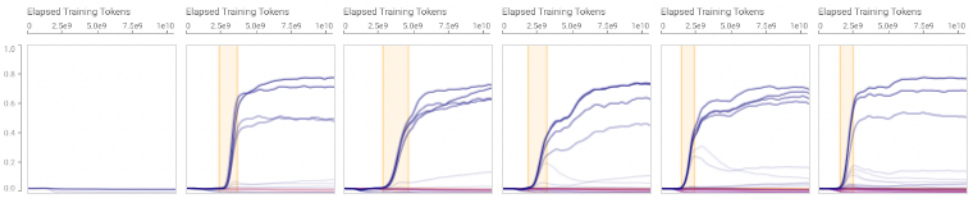
learning. More...

ATTENTION HEADS OVER TRAINING

"PREFIX MATCHING SCORE"

Each line is an attention head. Y axis is the average fraction of a head's attention weight given to the token we expect an induction head to attend to -- the token where the prefix matches the present context -- on random, synthetic data. Heads are colored by layer depth: early; late. More...



TRACE OF QK EIGENVALUES

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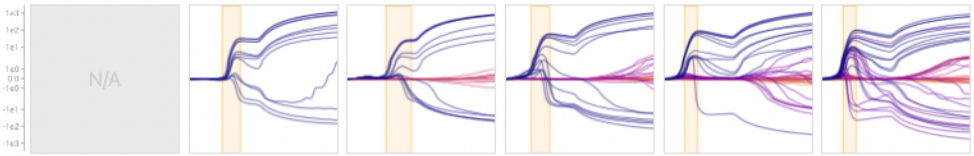

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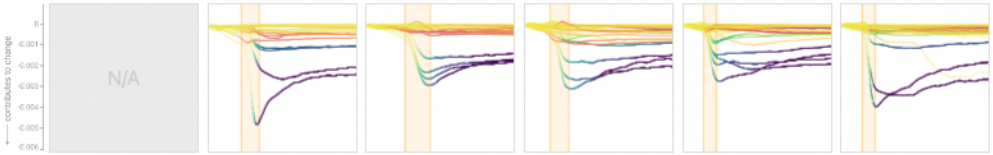

ABLATION ATTRIBUTION TO "IN-CONTEXT LEARNING SCORE"

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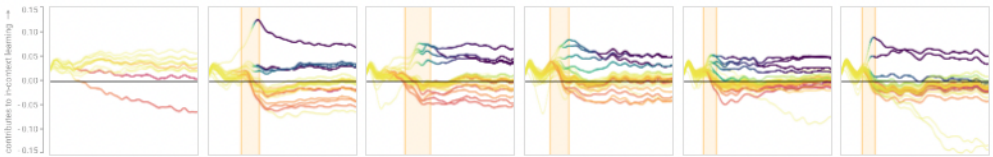


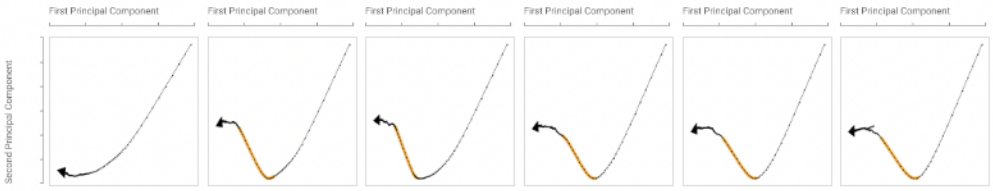
SMALL TRANSFORMERS (WITH MLPS)

This experiment series studies small normal transformers (with MLP layers, unlike the above). For these small models, we're able to capture snapshots with a high time resolution over early training. The main observation to take away is that the results are essentially the same as the attention-only models (where we have a stronger theoretical understanding. More...

ONE LAYER (WITH MLPS) | TWO LAYER (WITH MLPS) | THREE LAYER (WITH MLPS) | FOUR LAYER (WITH MLPS) | FIVE LAYER (WITH MLPS) | SIX LAYER (WITH MLPS)

PCA OF TOKEN LOSSES

The vector of per-token losses is a way to map different neural network behavior to the same vector space. We take 10,000 individual token predictions per model, and project them onto the first two principal components. This shows how the large-scale-behavior of multiple networks evolve over training. More...

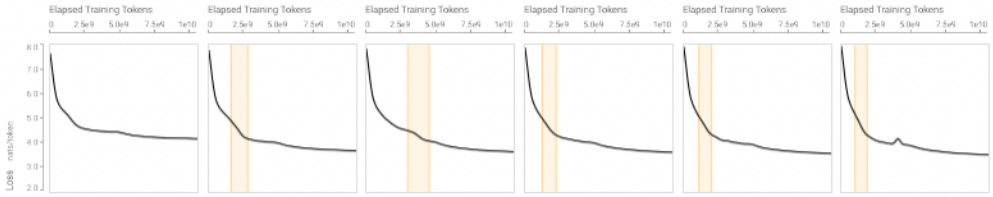


MODELS OVER TRAINING

LOSS OVER TRAINING

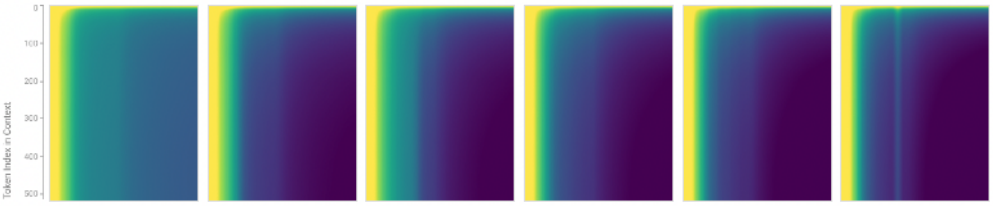
Loss curve showing $\mathrm{Loss}(n_{\mathrm{train}})$ after the model is trained on n_{train} tokens of training data. More...

The orange band denotes phase change interval.



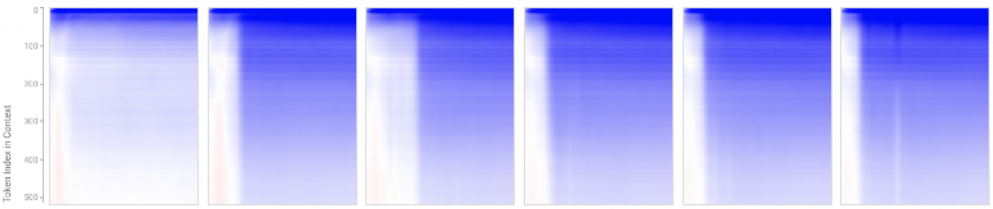
LOSS OVER TRAINING, BROKEN DOWN BY CONTEXT INDEX

Heatmap of $\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}})$, the average loss of i_{ctx} token in context after n_{train} elapsed tokens of training. More...



DERIVATIVE OF LOSS WRT CONTEXT INDEX

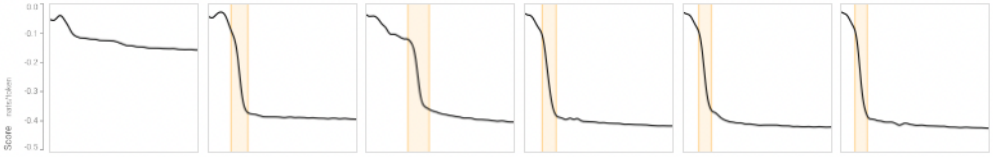
Heatmap of $d\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}})\ /\ d\ln(i_{\mathrm{ctx}})$. The log vertical partial derivative of the above graph. More...



"IN-CONTEXT LEARNING SCORE"

$\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}} = 500)$
$- \mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}} = 50))$

How much better the model is at predicting the 500th token than the 50th? We use this as a proxy for how good the model is at in-context learning. More...



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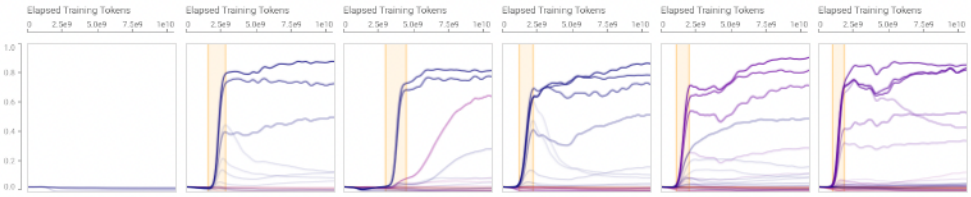

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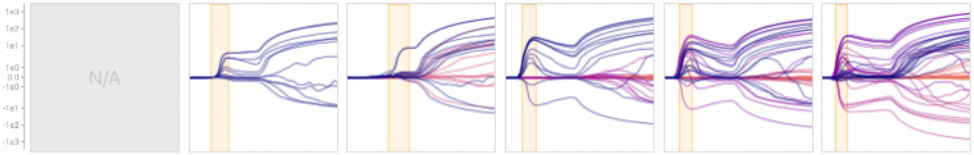

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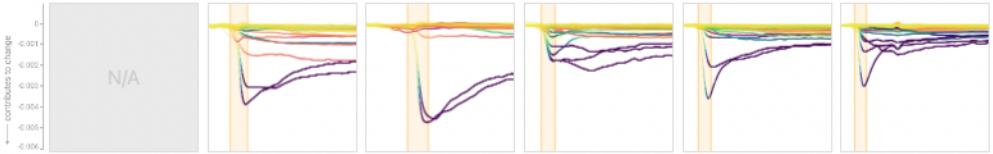

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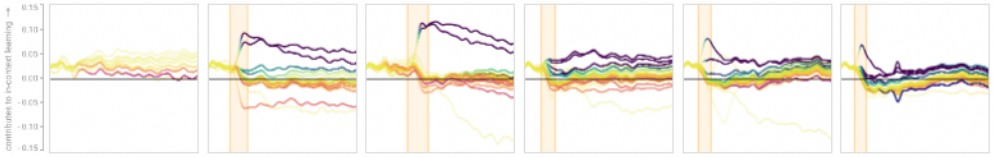
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